Why do apps like Instagram, Amazon, and Netflix feel insanely fast... even with millions of users? Are they hitting the database every single time?

What is Caching?

- Temporary storage for frequently accessed data
- Reduces direct database calls
- Improves application speed and performance

Why Caching is Important?

- Faster response times
- Reduces database load
- Handles high traffic efficiently
- Improves scalability

What is Redis?

- In-memory data store
- Extremely fast compared to databases
- Commonly used for caching in distributed systems

What is Cache Aside Pattern?

- Application checks cache first
- If data exists → return from cache
- If not → fetch from database
- Store data in cache for future requests

Cache Aside Flow

- Request comes in
- Check Redis cache
- Cache hit → fast response
- Cache miss → query database
- Save result in cache

What is Write-Through Pattern?

- Data written to database and cache together
- Cache always stays updated
- Prevents stale/outdated data

Techzeen

Write-Through Flow

- User updates data
- Application updates database
- Cache updated immediately
- Future reads become faster

Cache Aside vs Write-Through

Real-World Use Cases

Techzeen

Important Caching Concepts

Why Redis is Popular in Microservices